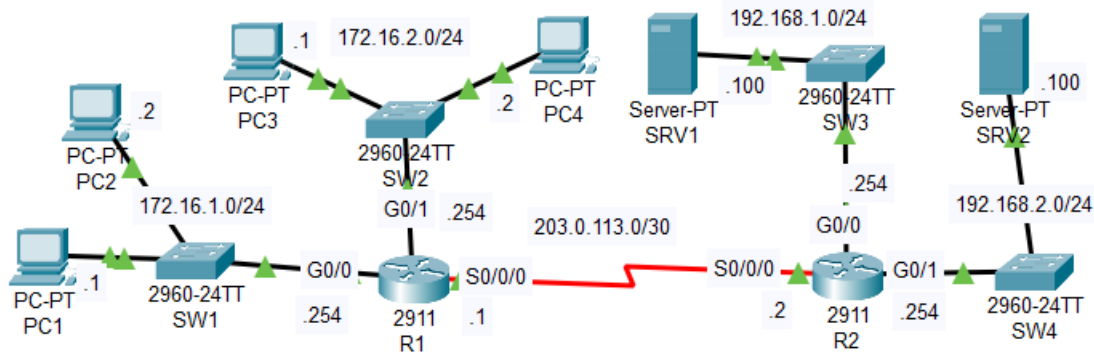




Topology

1. Configure OSPF on R1 and R2 to allow full connectivity between the PCs and servers.

```
R1(config)#router ospf 1
R1(config-router)#network 172.16.0.0 0.0.255.255 area 0
R1(config-router)#network 203.0.113.0 0.0.0.3 area 0
R1(config-router)#do sh ip ospf int
```

GigabitEthernet0/0 is up, line protocol is up
 Internet address is 172.16.1.254/24, Area 0
 Process ID 1, Router ID 203.0.113.1, Network Type BROADCAST, Cost: 1
 Transmit Delay is 1 sec, State DR, Priority 1
 Designated Router (ID) 203.0.113.1, Interface address 172.16.1.254
 No backup designated router on this network
 Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
 Hello due in 00:00:05
 Index 1/1, flood queue length 0
 Next 0x0(0)/0x0(0)
 Last flood scan length is 1, maximum is 1
 Last flood scan time is 0 msec, maximum is 0 msec
 Neighbor Count is 0, Adjacent neighbor count is 0
 Suppress hello for 0 neighbor(s)

GigabitEthernet0/1 is up, line protocol is up
 Internet address is 172.16.2.254/24, Area 0
 Process ID 1, Router ID 203.0.113.1, Network Type BROADCAST, Cost: 1
 Transmit Delay is 1 sec, State DR, Priority 1
 Designated Router (ID) 203.0.113.1, Interface address 172.16.2.254
 No backup designated router on this network
 Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
 Hello due in 00:00:05
 Index 2/2, flood queue length 0
 Next 0x0(0)/0x0(0)
 Last flood scan length is 1, maximum is 1
 Last flood scan time is 0 msec, maximum is 0 msec
 Neighbor Count is 0, Adjacent neighbor count is 0
 Suppress hello for 0 neighbor(s)

Serial0/0/0 is up, line protocol is up
 Internet address is 203.0.113.1/30, Area 0
 Process ID 1, Router ID 203.0.113.1, Network Type POINT-TO-POINT, Cost: 64
 Transmit Delay is 1 sec, State POINT-TO-POINT,
 Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
 Hello due in 00:00:03
 Index 3/3, flood queue length 0
 Next 0x0(0)/0x0(0)

```
R2(config)#router ospf 1
R2(config-router)#network 192.168.0.0 0.0.255.255 area 0
R2(config-router)#network 203.0.113.0 0.0.0.3 area 0
R2(config-router)#do sh ip ospf int

GigabitEthernet0/0 is up, line protocol is up
  Internet address is 192.168.1.254/24, Area 0
  Process ID 1, Router ID 203.0.113.2, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 203.0.113.2, Interface address 192.168.1.254
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:03
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
GigabitEthernet0/1 is up, line protocol is up
  Internet address is 192.168.2.254/24, Area 0
  Process ID 1, Router ID 203.0.113.2, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 203.0.113.2, Interface address 192.168.2.254
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
--More--
23:14:40: %OSPF-5-ADJCHG: Process 1, Nbr 203.0.113.1 on Serial0/0/0 from LOADING to FULL, Loading
Done
  Hello due in 00:00:03
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/0/0 is up, line protocol is up
  Internet address is 203.0.113.2/30, Area 0
  Process ID 1, Router ID 203.0.113.2, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured. Hello 10. Dead 40. Wait 40. Retransmit 5
```

```
R2(config-router)#do sh ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
203.0.113.1	0	FULL/ -	00:00:31	203.0.113.1	Serial0/0/0

```
R2(config-router)#do sh ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

```
172.16.0.0/24 is subnetted, 2 subnets
O    172.16.1.0/24 [110/65] via 203.0.113.1, 00:01:45, Serial0/0/0
O    172.16.2.0/24 [110/65] via 203.0.113.1, 00:01:45, Serial0/0/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, GigabitEthernet0/0
L    192.168.1.254/32 is directly connected, GigabitEthernet0/0
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/24 is directly connected, GigabitEthernet0/1
L    192.168.2.254/32 is directly connected, GigabitEthernet0/1
203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C    203.0.113.0/30 is directly connected, Serial0/0/0
L    203.0.113.2/32 is directly connected, Serial0/0/0
```

```
R1(config-router)#do sh ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

```
172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
C    172.16.1.0/24 is directly connected, GigabitEthernet0/0
L    172.16.1.254/32 is directly connected, GigabitEthernet0/0
C    172.16.2.0/24 is directly connected, GigabitEthernet0/1
L    172.16.2.254/32 is directly connected, GigabitEthernet0/1
O    192.168.1.0/24 [110/65] via 203.0.113.2, 00:02:38, Serial0/0/0
O    192.168.2.0/24 [110/65] via 203.0.113.2, 00:02:38, Serial0/0/0
203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C    203.0.113.0/30 is directly connected, Serial0/0/0
L    203.0.113.1/32 is directly connected, Serial0/0/0
```

2. Configure standard numbered ACLs on R1 and standard named ACLs on R2 to fulfill the following network policies:

- Only PC1 and PC3 can access 192.168.1.0/24
- Hosts in 172.16.2.0/24 can't access 192.168.2.0/24
- 172.16.1.0/24 can't access 172.16.2.0/24
- 172.16.2.0/24 can't access 172.16.1.0/24

```
R2(config)#ip access-list standard TO_192.168.1.0/24
R2(config-std-nacl)#permit 172.16.1.1
R2(config-std-nacl)#permit 172.16.2.1
R2(config-std-nacl)#deny any
R2(config-std-nacl)#int g0/0
R2(config-if)#ip access-group TO_192.168.1.0/24 out
R2(config-if)#ip access-list standard TO_192.168.2.0/2
R2(config-std-nacl)#deny 172.16.2.0 0.0.0.255
R2(config-std-nacl)#permit any
R2(config-std-nacl)#int g0/1
R2(config-if)#ip access-group TO 192.168.2.0/24 out

R1(config)#access-list 1 deny 172.16.1.0 0.0.0.255
R1(config)#access-list 1 permit any
R1(config)#access-list 2 deny 172.16.2.0 0.0.0.255
R1(config)#access-list 2 permit any
R1(config)#int g0/1
R1(config-if)#ip access-group 1 out
R1(config-if)#int g0/0
R1(config-if)#ip access-group 2 out
```